CSCI 620: Introduction to Big Data, Fall 2261

Project description

The goal of this project is for students to practice the concepts studied in the course by exploiting different datasets. The project consists of three phases:

Phase I

Select one or more datasets to be used in the rest of the project. You may use existing datasets, or may compile your own. The final dataset needs to be large (~50M tuples in a relational database), and interesting enough so you can perform meaningful queries and mine meaningful information from it. You need to provide a link to the dataset, a detailed description of the data, a meaningful relational model to faithfully represent the dataset, and a program to load the dataset. Some suggestions on where to find datasets are given in the Resources section of myCourses. **Do not include your dataset in your submission.**

Phase II

a. Provide a program that issues at least five interesting SQL queries over the previous relational model and propose and explain indexes to speed up query execution. Report your timings before and after indexing. Discover and report functional dependencies along with an explanation for these dependencies. Discuss normalization with respect to the relational model you provided in Phase I.

b. Propose a document-oriented model for your dataset and compare it with your relational model. You should aim to make effective use of the document model as discussed in class. This means your model will likely be different from your relational model. You should also provide code to load your data into this model. Your report should include details on the structure of each collection you create.

Phase III

a. Implement a suite of 5 complex MongoDB aggregation queries that leverage document capabilities such as embedded arrays, sub-documents, and pipeline operators. Include at least one multi-stage aggregation using $match, $group, $project, and $sort. Include at least one relational-style join using $lookup to join secondary collections (or explain why embedding eliminated the need for it based on your Phase II model). Perform array

manipulation (e.g., using $unwind, $arrayElemAt, or $filter) or geospatial/text queries if applicable to your dataset.

b. Compare the execution performance of equivalent workload queries between Relational (Phase II SQL) and Document (Phase III MongoDB) models.

Final presentations

Prepare a presentation of your project approximately 15-20 minutes in length. You have complete freedom for this task about format and contents: You need to make sure to talk about the inner details and concepts, but you also need to make it attractive to other students. You should aim to explain your dataset as well as the specific insights you found while completing each phase of your project. Note that it is not necessary to explain concepts taught in class since your classmates should already know this material.